Learning Objective 3.4: I can calculate the phase weights required to steer a beam to a given angle and predict the resulting array pattern.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From geometry to phase.** The course array is a uniform linear array of $N = 8$ patch elements spaced $d = 14$ mm apart, receiving at $f = 10.3$ GHz. A plane wave arrives $\theta_0 = 25^\circ$ from broadside.

   (a) In one or two sentences, explain which element the wavefront reaches first and why a fixed phase applied to each element can make all eight received signals add together.

   The wave arrives from the $+25^\circ$ side, so it reaches the highest-numbered element first and each lower-numbered element a little later. Successive elements therefore see the same signal with a fixed phase difference between neighbors. Adding a compensating phase to each element cancels that difference, so all eight signals arrive at the summing junction in phase and add coherently.

   (b) Compute the wavelength, the electrical spacing d/λ, and the product kd in degrees.

   $$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8}{10.3 \times 10^9} = 0.02911 \text{ m} = 29.11 \text{ mm}$$
   $$\frac{d}{\lambda} = \frac{14}{29.11} = 0.481$$
   $$kd = 2\pi\,(0.481) = 3.02 \text{ rad} = 173.2^\circ$$

   kd = 173.2°

   (c) Compute the progressive element-to-element phase $\Delta\phi$ required to steer the beam to 25°.

   $$\Delta\phi = kd\sin(\theta_0) = 173.2^\circ \times \sin(25^\circ)$$
   $$= 173.2^\circ \times 0.4226 = 73.2^\circ$$

   $\Delta\phi$ = 73.2°

   (d) Compute the arrival-time difference between neighboring elements at this steer angle, in picoseconds.

   $$\Delta t = \frac{d\sin(\theta_0)}{c} = \frac{(0.014)\,(0.4226)}{2.998 \times 10^8}$$
   $$= 1.97 \times 10^{-11} \text{ s} = 19.7 \text{ ps}$$

   Δt = 19.7 ps

2. **The phase table.** We command the same array to $\theta_0 = -40^\circ$. Element 0 is the phase reference, and the ADAR1000 accepts a phase setting between 0° and 360°.

   (a) Compute $\Delta\phi$ for this steer angle, including its sign.

   $$\Delta\phi = 173.2^\circ \times \sin(-40^\circ)$$
   $$= 173.2^\circ \times (-0.6428) = -111.3^\circ$$

   $\Delta\phi$ = -111.3°

   (b) Compute the commanded phase of element 5 from $\phi_n = n\Delta\phi$, before any wrapping.

   $$\phi_5 = 5\,(-111.3^\circ) = -556.5^\circ$$

   ϕ_5 = -556.5°

   (c) Give the value written to element 5 after wrapping into 0° to 360°.

   $$\phi_5 = -556.5^\circ + 720^\circ = 163.5^\circ$$

   ϕ_5 = 163.5°

   Two whole turns are added.

   (d) Wrapping changed the number by two whole turns and did not change the pattern. Explain why in one or two sentences, and state what information the wrapped table no longer carries.

   A phase difference of 360° is one full cycle of the carrier, so the element radiates exactly the same field either way; at a single frequency the two settings are indistinguishable. What is lost is the record of the true time delay: the unwrapped ramp corresponds to a specific delay in seconds, while the wrapped values only reproduce that delay at the design frequency. That is why a wideband array needs time-delay units rather than phase shifters alone.

3. **The steered pattern.** The array is steered to $\theta_0 = 50°$. Use $\psi = kd\,(\sin(\theta) - \sin(\theta_0))$ with $kd = 173.2°$, and recall $\lambda/Nd = 29.1/112 = 0.260$.

   (a) Compute the array-factor argument ψ at the observation angle $\theta = 30°$.

   $\psi =$ $-46.1°$

   $$\psi = 173.2°\,(\sin(30°) - \sin(50°))$$
   $$= 173.2°\,(0.500 - 0.766) = -46.1°$$

   (b) Locate the first null on the broadside side of the main lobe from $\sin(\theta) = \sin(\theta_0) - \lambda/Nd$.

   $\theta =$ $30.4°$

   $$\sin(\theta) = 0.766 - 0.260 = 0.506$$
   $$\theta = \arcsin(0.506) = 30.4°$$

   The main lobe therefore reaches $19.6°$ below its peak.

   (c) The matching null on the other side requires $\sin(\theta) = 0.766 + 0.260 = 1.026$. Explain what that result means physically and what the pattern looks like on that side of the beam.

   No real angle has a sine greater than one, so that null lies outside the visible region and never appears in the pattern. On that side the main lobe falls without a null, all the way to -19.0 dB at $90°$, and never rises again into a sidelobe. The beam is asymmetric about $50°$ and has no sidelobe on its endfire side.

   (d) Two commanded angles differ by $5°$ near broadside, and two others differ by $5°$ near $60°$. Which pair requires the larger change in $\Delta\phi$, and what does that mean for an array searching a sector in equal phase steps?

   The pattern is rigid in $\sin(\theta)$, and $\Delta\phi$ is proportional to $\sin(\theta_0)$, so a $5°$ move near broadside changes $\sin(\theta_0)$ much more than the same move near $60°$: the pair near broadside needs the larger phase change: $0°$ to $5°$ needs $15.1°$, while $55°$ to $60°$ needs only $8.1°$. An array stepped in equal increments of $\Delta\phi$ therefore takes small angular steps near broadside and large ones near endfire, so a uniform search grid in angle requires unequal phase steps.

4. **Beam broadening and scan loss.** Use $\theta_{\text{HP}} \approx 0.886\lambda/\left(Nd\cos\left(\theta_0\right)\right)$ and the broadside uniform-array directivity $D \approx 2Nd/\lambda$, continuing with $N = 8$, $d = 14$ mm, and $\lambda = 29.1$ mm.

   (a) Compute the broadside half-power beamwidth of the array.

$$\theta_{\text{HP}}\left(0\right) = \frac{0.886\left(29.1\right)}{112} = 0.2302 \text{ rad}$$
$$= 13.2^\circ$$

   This is the canonical broadside beamwidth of the course array.

θ_{HP} = 13.2°

   (b) Compute the half-power beamwidth when the beam is steered to $\theta_0 = 55^\circ$.

$$\theta_{\text{HP}}\left(55^\circ\right) = \frac{13.2^\circ}{\cos\left(55^\circ\right)} = \frac{13.2^\circ}{0.574}$$
$$= 23.0^\circ$$

θ_{HP} = 23.0°

   (c) Compute the broadside directivity of the array in dBi. Then, using the fact that the effective aperture shrinks to its projection $Nd\cos\left(\theta_0\right)$ when the beam is steered, compute the expected gain reduction at $\theta_0 = 55^\circ$ and name the effect.

$$D\left(0\right) = \frac{2\left(112\right)}{29.1} = 7.70 \rightarrow 10\log_{10}\left(7.70\right) = 8.9 \text{ dBi}$$
$$\Delta G = 10\log_{10}\left(\cos\left(55^\circ\right)\right)$$
$$= 10\log_{10}\left(0.574\right) = -2.4 \text{ dB}$$

   This is **scan loss**, carried by the element pattern. The array's beam is a cone about its axis: steering shrinks the cone's circumference by $\cos\left(\theta_0\right)$ as the beam widens by $1/\cos\left(\theta_0\right)$, so the isotropic array factor keeps its directivity (7.8 at 55°). Each element's projected area shrinks as $\cos\left(\theta\right)$ (Lesson 21).

$D\left(0\right)$ = 8.9 dBi

ΔG = -2.4 dB

   (d) A tracking requirement allows the beam to broaden to at most 20°. Find the largest usable scan angle, and state the trade-off this puts on the system design in one sentence.

$$\cos\left(\theta_0\right) \geqslant \frac{13.2^\circ}{20^\circ} = 0.66$$
$$\theta_0 \leqslant \arccos\left(0.66\right) = 48.7^\circ$$

   A wider field of view at this beamwidth requires a longer aperture: only an array with more elements keeps $\theta_{\text{HP}} \leqslant 20^\circ$ after its projected length shrinks by $\cos\left(\theta_0\right)$.

θ_0 = 48.7°

5. **The inverse problem.** You read the following phase settings out of the beamformer. The array is the course array, $kd = 173.2°$.

n	0	1	2	3	4	5	6	7
ϕ_n (deg)	0	260.7	161.4	62.1	322.8	223.5	124.2	24.9

(a) Difference neighboring elements, unwrap, and report the common step and the value of $\Delta\phi$.

$\Delta\phi =$ $-99.3°$

$$\phi_2 - \phi_1 = 161.4° - 260.7° = -99.3°$$
$$\phi_1 - \phi_0 = 260.7° \rightarrow 260.7° - 360° = -99.3°$$

All seven differences reduce to $-99.3°$, and the ramp is $\phi_n = n\Delta\phi$, so $\Delta\phi = -99.3°$.

(b) Recover the commanded steer angle.

$\theta_0 =$ $-35.0°$

$$\sin(\theta_0) = \frac{\Delta\phi}{kd} = \frac{-99.3°}{173.2°} = -0.573$$
$$\theta_0 = \arcsin(-0.573) = -35.0°$$

A falling ramp means a negative $\Delta\phi$ and a beam on the negative side of broadside.

(c) Suppose the seven differences had come out as $-99.3°$ for six of them and $-84°$ for one. State two things that could produce that result, and what you would check first.

The array is not carrying a single uniform steering ramp, and where the odd difference sits narrows the cause. An offset δ on an interior element would spoil two adjacent differences, one by $+\delta$ and the next by $-\delta$ (here $+15.3°$ and $-15.3°$). A single odd difference therefore means either an offset on an end element (element 0 or 7), or, if it is $\phi_4 - \phi_3$, a phase error between the two ADAR1000 subarray channels, which moves elements 4 to 7 together. A taper affects amplitude and not phase, so it is not a candidate. Check which difference is the odd one first: if it is $\phi_4 - \phi_3$, rerun Calibrate; otherwise check the calibration offset of that end element.

(d) The ADAR1000 sets phase on a $2.8125°$ grid. Give the phase element 1 takes on that grid and the resulting error.

$\phi_1 =$ $261.6°$

$$\frac{260.7°}{2.8125°} = 92.69 \rightarrow 93 \text{ steps}$$
$$\phi_1 = 93\,(2.8125°) = 261.6°$$

The error is $+0.9°$, which is within half a step as expected. L23 computes how much a grid of this size raises the sidelobe level and reduces null depth.

Documentation: _____